



NTRIP Client Devices / Best Practices

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Introduction

This document seeks to provide general guidance for developers of NTRIP Client devices. The design guidance expressed here represents a set of Best Practices that the designer of every NTRIP Client device should follow to avoid connection problems with the NTRIP Caster. Failure to follow these recommendations can result in a failure to connect to the desired resource (GNSS Base Station data stream) and in some cases the NTRIP Caster taking action to block the connecting device and/or its IP from access to the Caster. All requirements and recommendations within this document are made to promote best practices and are not designed to supersede the published standards.

The RTCM SC-104 NTRIP standard provides two protocol variations, both are considered here. These are often referred to as either the “Version 1 style” or “Version 2 style” or as the “Rev 1” or “Rev 2” style. The terms *revision* (“Rev” for short) or *version* may be considered the same and both are found in common use. It is strongly preferred that all connections now use the Version 2 style whenever possible. This document provides examples for both.

The NTRIP standard uses a small subset of the http protocol. Specifically, for NTRIP Clients; the **GET** verb is used to request a resource from the NTRIP Caster (the server element of the system). A similar process is used to send¹ data from a base station. NTRIP Clients **do not** need to implement all the various verbs defined in http and NTRIP Client developers should not presume that NTRIP Casters can process any http verbs or meta data not expressly defined in the NTRIP standard itself.

The NTRIP standard follows the common http protocol practice that every line in the request ends with <CR><LF>. It also follows the http practice of ending the request with <CR><LF><CR><LF>. This final “blank” line indicates that the request has completed. The header lines within the request can appear in any order and may contain additional meta data lines that may be ignored by the Caster. However, this is not normally an issue as NTRIP connections are typically fairly terse.

A minimal well-formed example of a Version 2 NTRIP Client connection is as follows:

```
GET /MountPtName HTTP/1.1<CR><LF>
Host: Acaster.com:2101<CR><LF>
Ntrip-Version: Ntrip/2.0<CR><LF>
User-Agent: NTRIP ProductName/Version<CR><LF>
<CR><LF>
```

A minimal well-formed example of the older (now outdated) Version 1 NTRIP Client connection is as follows:

```
GET /MountPtName HTTP/1.0<CR><LF>
Host: Acaster.com:2101<CR><LF>
User-Agent: NTRIP ProductName/Version<CR><LF>
<CR><LF>
```

¹ When sending data the POST verb is used (in NTRIP Version 2). In NTRIP Version 1 the verb “SOURCE” was invented and used for this purpose. Other protocol changes were also made.

In the above the text `"MountPtName"` represents the GNSS Base Station mount point name (the mountpoint). This is the name of the Base Station data stream that is being requested. In generic http terms, this is the requested resource. It is important to note that Mount point names are **ALWAYS** case sensitive and consist of the letters A~Z, a~z, the digits 0~9, the dash ("-") and underscore ("_") characters **ONLY**. Therefore "MyBase" and "mybase" are considered two distinct different mountpoints, while "My Base" is not valid due to the space. Specific NTRIP Casters may have other restrictions on naming (such as starting with a letter). Generally, GNSS Base Stations are assigned mountpoint names that are short and well under 30 characters, although the standard does allow longer names, up to 100 characters. The returned Caster sourcetable serves to provide a list of the valid mount point names from which an NTRIP Client may select.

In the above the text `"Acaster.com:2101"` represents the DNS name of the NTRIP Caster and the port on which it is to be found. This line should **ALWAYS** be present. This line is NOT case sensitive, following the internet rules used for URLs and domain name. It is discussed in more depth in a moment.

In the above the text `"ProductName/Version"` represents the name of the NTRIP Client software (and any release version details) being used to connect to the NTRIP Caster. This is a self-assigned string created and maintained by the software developer consisting only of ASCII characters. This line should **ALWAYS** be present. The preceding "NTRIP" keyword, is required. The normal convention is to send this line as `"User-Agent : NTRIP ProductName/Version"` however the standard also allows additional text to be present on the same line after another space. Developers of NTRIP Client software should select a string which avoids infringement on other firms product names, trademarks, service marks, and registered names. Some NTRIP Casters will modify their response based on this line, allowing them to better serve different types of users. Also of note: many NTRIP Casters may presume the request comes from a browser rather than an NTRIP Client when this line is not present.

In the above text, and in http and NTRIP in general, a single space (hex 0x20) is used to separate the tokens in a line. Both http and NTRIP are rather strict about this point. Having additional "white space" present can lead to connection failures. Some NTRIP Casters are more tolerant of this than others. Particular care should be taken to avoid allowing a GUI design to insert unwanted spaces and thereby create invalid input values.

The above minimal examples do not present any additional lines of information in the request, although such lines can be present. This is a general design principle of http extensibility. That is, any request may contain additional lines (each ending with <CR><LF>) which are considered to be part of the request. Often this is used to provide details about the requestor's abilities or preferences for returned data. These lines may appear in any order. Intermediary http processing may also add or remove such lines. For the needs of developing simple NTRIP Client requests, only one additional line is needed: the NMEA-0183 \$GGA request. This is used when the NTRIP Client must tell the NTRIP Caster its approximate location. This use case is considered further in the below text.

On Sending the User / Password

Neither of the above examples provide any means for sending a user account name and account password. A prior user account on the Caster is commonly required² to access the resources of an NTRIP Caster. The assignment of the user account details is beyond the scope of this document. Most NTRIP Caster operators have some form of registration process that, upon completion, then results in a unique assigned user account name and password. NTRIP uses the common Base64 encoding method to represent the user and password separated by the char “:”. For example: the text “user:password” is then transmitted in the examples that follow as “dXNlcjpwYXNzd29yZA==”.

A typical well-formed example of a Version 2 NTRIP Client connection is as follows:

```
GET /MountPtName HTTP/1.1<CR><LF>
Host: Acaster.com:2101<CR><LF>
Ntrip-Version: Ntrip/2.0<CR><LF>
User-Agent: NTRIP ProductName/Version<CR><LF>
Accept: */*<CR><LF>
Connection: close<CR><LF>
Authorization: Basic dXNlcjpwYXNzd29yZA==<CR><LF>
<CR><LF>
```

A typical well-formed example of the older (now outdated) Version 1 NTRIP Client connection is as follows:

```
GET /MountPtName HTTP/1.0<CR><LF>
Host: Acaster.com:2101<CR><LF>
User-Agent: NTRIP ProductName/Version<CR><LF>
Accept: */*<CR><LF>
Connection: close<CR><LF>
Authorization: Basic dXNlcjpwYXNzd29yZA==<CR><LF>
<CR><LF>
```

NTRIP Client software that implements the above, as well as the various connection rules that follow, will be able to successfully connect to any NTRIP Caster which follows the standards. The actual RTCM NTRIP standard should be consulted for further details and options.

² Very few NTRIP Casters are operated where no user account is required to access the Base Station resources. These “open” Casters do not require a user name and password, so the line may be omitted.

Connecting to the NTRIP Caster

This section covers key design considerations when the NTRIP Client opens a socket and attempts to connect to a remote NTRIP Caster using TCP/IP.

Consider a successful connection by an NTRIP Client. Upon a successful connection, the NTRIP Caster replies by sending one of the following.

A minimal well-formed example of a Version 2 NTRIP Caster response is as follows:

```
HTTP/1.1 200 OK<CR><LF>
```

A minimal well-formed example of the older (now outdated) Version 1 NTRIP Caster response³ is as follows:

```
ICY 200 OK<CR><LF>
```

After this reply, the NTRIP Caster sends data from the selected Base Station stream back over the socket. No further NTRIP protocol data is sent by the NTRIP Caster. Typically, no further data is sent by the NTRIP Client either, with the exception of periodic NMEA-0183 \$GGA sentences. [The topic of sending NMEA-0183 \$GGA to the Caster is covered below.]

In practice the reply sent by a Version 2 NTRIP Caster will often contain additional http meta data about the Caster and may enumerate various supported features. A typical example would be as follows:

```
HTTP/1.1 200 OK<CR><LF>
Date: Wed, 11 November 2020 20:29:36 UTC<CR><LF>
Server: SNIP simpleNTRIP_Caster_[3.xx.00]<CR><LF>
Ntrip-Version: Ntrip/2.0<CR><LF>
Ntrip-Flags: <CR><LF>
Cache-Control: no-store, no-cache, max-age=0<CR><LF>
Pragma: no-cache<CR><LF>
Connection: close<CR><LF>
Content-Type: gnss/data<CR><LF><CR><LF>
```

The reply sent by a Version 2 NTRIP Caster may also contain the `Ntrip-Flags:` line with any additional data regarding any Version 2 features (such as Caster sourcetable filtering) that the Caster supports.

The reply ends with the chars `<CR><LF><CR><LF>` being sent, the empty line signifying the end of the http header. [Note also that in some Version 1 NTRIP Casters the final empty line is not present.]

The reply defined by Version 2 also defines a fallback mechanism for older Casters. A Version 2 NTRIP Client should expect that it *could* also get an NTRIP Version 1 response when sending an NTRIP Version 2 request to a Version 1 NTRIP Caster. The absence of the `Ntrip-Version` keyword in the header indicates a Version 1 NTRIP Caster reply.

³ NTRIP Version 1 was modeled after an audio streaming protocol called SHOUTcast, developed by NULLsoft/WinAmp, see <https://en.wikipedia.org/wiki/SHOUTcast> The term “ICY” comes from “I can Hear You” which is used there.

In an ideal world, the connection request from the NTRIP Client is accepted by the NTRIP Caster, an initial reply is sent back as defined above, and then a stream of data (typically RTCM 3 message content⁴) is returned to the NTRIP Client.

When this does not occur a number of best practices come into play to avoid abuse of the NTRIP Caster and to ensure the optimum performance of the NTRIP Client. The general goal of these rules is to optimize the NTRIP Client performance without creating excessive network traffic or loads on the NTRIP Caster. Developers of NTRIP Clients should confirm conformance with the following four points.

- An NTRIP Client should **NEVER** reconnect to an NTRIP Caster more often than once a second.

These are TCP/IP connections, not UDP, so they are never “lost” in transit. If there is a network connectivity problem (covered in the next section) it may take several seconds to become evident. Repeating failed connections from NTRIP Clients to the NTRIP Caster has become a major problem for Caster operators. As a general observation, if the NTRIP Caster replies with an error condition of some sort, the NTRIP Client should NOT reconnect again before addressing it. Use should be made of the returned error message!

- When an NTRIP Client is making reconnections to test for the presence of a Base Station mountpoint after a prior absence, the time period (the delay) between reconnections should increase proportionally with the number of prior failed attempts made.

A decreasing reconnection rate is recommended. As an example, an NTRIP Client might reconnect every 1,2,3,3,3,5,5,10,10,15,15,30th seconds after a failure, and slower thereafter. After an extended period of connection failures, a reconnection rate of not more than once every ~5 minutes should be used. An NTRIP Client should **NEVER** reconnect more than a dozen times over any 60 second interval.

- An NTRIP Client should **NEVER** connect unless it is ready to operate.

Some devices will connect to the NTRIP Caster and then disconnect when data is returned due to internal issues. Some devices will connect to a mountpoint on the NTRIP Caster which requires a \$GGA sentence with their location when they do not yet have a suitable location estimate.

Devices which repeatedly connect / disconnect to the NTRIP Caster for very short periods of time may be banned by the NTRIP Caster. This condition can also be caused by poor TCP/IP connections (often found on cellular networks in fringe coverage areas) and the NTRIP Caster is typically unable to determine the root cause.

Aside: In a similar fashion, some NTRIP Servers will connect to the NTRIP Caster when they have no actual available data to send into the Caster. This should be prevented.

⁴ NTRIP supports sending any type of data over the connection once it is established. While binary encoded RTCM 3.x message content is the most common, many proprietary data formats are also in use.

- An NTRIP Client should **ALWAYS** process the returned data on a connection failure.

In the case of a Version 2 reply, the 4xx error messages provide suitable insights into the nature of the problem. This should be processed and, as required, changes made to the next request before being sent. See the sections labeled **On Caster responses** and **On Standard Caster Error messages** for further guidance.

In the case of a Version 1 reply (when the Caster Source Table was returned) some processing can still be performed to determine the nature of the failure. For example, the table can easily be parsed to ensure that the requested mountpoint is present in the returned Caster Source Table.

As a general guideline, when a connection reaches the NTRIP Caster but does not result in success, the error reply should be examined, and corrections made rather than simply connecting again.

Network Issues

The above text deals with issues between the NTRIP Client and NTRIP Caster, chiefly conflicts with returning the requested resource (Base Station data stream). It ignores the general area of network connectivity between the two devices where other problems often occur. Connections between the NTRIP Caster and the NTRIP Client take place over TCP/IP using common industry standard communications methods which are well explained elsewhere. Consult any modern communications textbook for details. When these methods fail to connect, the response taken by the NTRIP Client can cause a variety of local problems to occur. As these problems are typically not seen by the NTRIP Caster, they are mentioned here only for completeness.

When connection to an NTRIP Caster is made by use of a Name (aka *example.com*), some form of DNS lookup service must be present to resolve the Name to an IP address (aka IP:1.2.3.4). Therefore, such a service must be provided at the NTRIP Client side, or alternatively the actual IP must be used to reach the Caster. The remote address must be reliably reachable by the device, or else network timeouts and packet dropouts will cause failure. Events which manifest themselves as timeouts to the Client (such as well-formed IP addresses that do not reply within a given time) are handled in the normal TCP/IP ways. Some Casters will also detect the “silly window syndrome” (in this context, sending only a few bytes of the NTRIP header in any TCP packet) and disconnect such users, presuming they are probing the Caster. Because TCP/IP is a robust and reliable network method, there is no need to rapidly start multiple new connections at service rates higher than once per second. Doing so may in fact overload, jam, and disrupt other local TCP/IP traffic in the same local network as the NTRIP Client device.

It should be observed that some Casters may use “chunking” (chunked transfer encoding, see RFC 9112) in the reply data stream.

How to Request a Caster Source Table

The proper way to request the Caster source table is with the “default” resource request “GET /” – this is true for both Version 1 and Version 2 connections. It is incorrect to use imaginary names such as “GET /castertable” for this. In a Version 1 Caster (or in a Version 2 Caster that sees a Version 1 request) the response to any request which makes no sense (such as asking for “castertable” the prior example) is to send back the entire Caster source table. This has the illusion⁵ of working for Version 1 connections.

The first line of an NTRIP Client table request connection is as follows:

```
GET / HTTP/1.1<CR><LF>
```

The remaining lines are the same as shown in prior examples. Requesting a Caster source table multiple times for extended periods without other operations (such as connecting to one of the mountpoints) may result in the NTRIP Caster blocking the connection.

Requests for a Caster source table should **ALWAYS** include the User Name and Password (in the `Authorization:` line). Failure to include this line may result in some Casters not returning a table.

Sending NMEA-0183 \$GGA to the Caster

In the NTRIP protocol use is made of the NMEA-0183 sentences to send location data about the NTRIP Client back to the NTRIP Caster. The indication that this information is *requested* or *required* by the Caster is a flag present in the Caster source table for each given mountpoint (see the standard for details). This information is **REQUIRED** for various NetworkRTK data streams in order to allow the NTRIP Caster to select the type of stream information to be returned. When an NTRIP Client connects to such a data stream and does NOT send any suitable NMEA-0183 sentences, the NTRIP Caster will typically drop the connection after a short period of waiting. At other times the information may simply be requested by the Caster, typically to track the gross location of connected users. The NTRIP standard and the Caster source table provide no indication regarding if the information is required or simply requested.

The NTRIP standard and the Caster source table provide no indication of how often the information should be updated once it is initially sent. A nominal time of from every 10th to every 30th second is suggested. Most NTRIP Casters provide a short period of time (tens of seconds or more) for the initial sentence to be sent after the connection is established, when it is required. The precise alignment to a fixed sending interval is not important. Some NTRIP Casters that create a synthetic base station for the user (a type of Network RTK stream) can require a longer period of time (perhaps 30 seconds) before the initial reply data becomes available. Following the connection points made above, the NTRIP Client should **NEVER** connect to the NTRIP Caster if it does not have a suitable⁶ and valid \$GPGGA sentence ready for transmission. Also, the NTRIP Client should not

⁵ If by chance there is a data stream matching the resource name, the request would be interpreted as trying to connect to that resource.

⁶ In other words, with values for latitude and longitude that are not zero/zero placeholders.

disconnect from the NTRIP Caster too quickly after the initial acknowledgement message (*200 OK* or *ICY 200*) is received while awaiting the start of stream data.

All NTRIP Clients should send the \$GPGGA sentence to the NTRIP Caster when the flag in the Caster source table for that mountpoint is set (to the value "1"). The sentence should be sent in the request header when using Version 2 and after the initial response reply when sending to a Version 1 Caster. The sentence should end with <CR><LF> at all times. This is true for the initial message as well as all subsequent messages sent. Well-formed examples⁷ follow below.

The \$GPGGA sentence in a Version 2 NTRIP Client request is sent in the header and is as follows:

Ntrip-GGA:

```
$GPGGA,110739.0,1220.70000000,N,06753.40600000,E,4,10,1.0,0.000,M,0.0,M,,*67<CR><LF>
```

The \$GPGGA sentence in the (now outdated) Version 1 NTRIP Client request is as follows:

```
$GPGGA,111123.0,1220.70000000,N,06753.40600000,E,4,10,1.0,0.000,M,0.0,M,,*6B<CR><LF>
```

And any subsequent updates in either version are as follows:

```
$GPGGA,111133.0,1220.70000000,N,06753.40600000,E,4,10,1.0,0.000,M,0.0,M,,*6A<CR><LF>
```

When an NTRIP Client requests a connection to a NetworkRTK data streams and provides a location which is outside the service area covered by the NetworkRTK data stream (the resource), an error may result. The Caster may reply with various 4xx error replies (if it is a Version 2 connection) or with the Caster source table (if it is a Version 1 connection). At this time the NTRIP protocol does not define more specific details about the error. Some NTRIP Casters do provide further details.

Questions occur about the difference between NMEA \$GPGGA and \$GNGGA sentences. In simple terms, the first two letters refer to the GNSS system used to produce the reply. The \$GPGGA sentence refers to a GPS based positioning solution, although it is commonly used for any solution. By contrast, the \$GNGGA sentence refers to the more generic "GNSS" indicating multiple systems were used. There are also sentences defined for the other major GNSS system (\$GL... = GLONASS, \$GA... = Galileo, \$GQ... = QZSS, \$GB/BD... = Compass/ BeiDou) although these are not seen in NTRIP use. For best compatibility it is recommended to only use the \$GPGGA sentence. However, some NTRIP Clients and some NTRIP Casters do support the use of \$GNGGA sentences.

A strictly compliant NMEA \$GGA sentence only allows 4 digits after the decimal point for minutes of latitude and longitude values (in the format DDMM.mmmm). Sending more is a violation of the NMEA specification but rarely causes an issue. The correct time stamp and checksum for the message should also be used, but some Casters are lax in checking this. A proper NMEA \$GGA sentence always ends with <CR><LF> and some Casters expect this. The location sent in the \$GGA sentence is not required to be RTK estimated location, it is often simply a least squares autonomous estimation. The resulting accuracy of a few meters is sufficient for any

⁷ The values used for time, latitude, and longitude in these examples are simply representative placeholders.

Network RTK needs. Some Casters require that the reported latitude and longitude values move and that the time remains correct.

While the NTRIP standard allows multiple NMEA sentences to be sent, it is recommended to ONLY send NMEA \$GGA sentences to the NTRIP Caster. Some NTRIP Casters will disconnect devices that send excessive NMEA content.

On Caster responses

A successful connection to a Version 2 NTRIP Caster results in a reply of “HTTP/1.1 200 OK<CR><LF>” (as well as other header details) followed by binary data (typically containing RTCM3 message content) as covered previously. From an OSI layer perspective, the NTRIP negotiation process takes place at the session layer while the flow of data is at the application layer.

An **unsuccessful** connection attempt results in an error code being returned, following by the TCP/IP socket being disconnected. The NTRIP Client device should determine why the connection failed (and what to change) **before** connecting again.

An NTRIP Client device that repeatedly connects to an NTRIP Caster with the same incorrect credentials or request is likely to be blocked or banned from connecting after a period of time. NTRIP Client devices that repeatedly connect to an NTRIP Caster with varying but incorrect credentials are also likely to be blocked or banned. The precise rules used will vary with the way each individual NTRIP Caster has been set up. The actual NTRIP standard is somewhat vague on this point, which allows NTRIP Caster developers and operators to implement different responses.

Most Version 2 NTRIP Casters return some form of an html-like 4xx error reply when an unsuccessful connection attempt occurs. See the next section for a summary of common errors. In NTRIP Version 2 we use some, but not all, of the well-known http response codes. The next section defines the ones used in NTRIP and provides further commentary regarding when they are used. The definitive guide to these codes can be found in RFC 9110 and its successors (see: <https://httpwg.org/specs/rfc9110.html#overview.of.status.codes>).

Aside: In NTRIP Version 1, no such codes are used. In the event of any type of connection problem (a non-existing mountpoint, bad user credentials, etc.), the complete Caster source table is returned. A returned Caster source table indicates there was a connection problem (unless the table was properly requested as defined elsewhere in this document). At the very least, the NTRIP Client should seek to confirm that the requested Base Station mountpoint is present in the returned table before attempting to reconnect again. Some Casters will also reply to poorly formed Version 1 requests with “ICY 401 Unauthorized<CR><LF>” as sort of a hybrid reply.

The returned http codes fall into five general categories:

- Informational responses (100–199) Not used or defined in NTRIP
- Successful responses (200–299) Only the ‘200’ code is used in NTRIP Version 2
- Redirection messages (300–399) Not yet used or defined in NTRIP, but may be found

- Client error responses (400–499) Some codes are defined in NTRIP Version 2
- Server error responses (500–599) Three codes are used in NTRIP Version 2

Although only those codes defined further in the next section are likely to be seen from an NTRIP Caster, a well-built NTRIP Client should be able to accept and process any of these. The most likely cause for seeing one of the unused codes would be a mistaken connection to a non-NTRIP endpoint. It should also be noted that different NTRIP Casters may reply with different error messaging depending on the level of informational leakage that the NTRIP Caster operator is willing to allow.

These general categories are handled in NTRIP as follows.

Information responses

NTRIP does not at this time support any of these.

Successful responses

When the NTRIP Caster accepts a connection request (the NTRIP Client GET request), it replies with the text “HTTP/1.1 200 OK<CR><LF>” regardless of the message format used in the data stream. Other common http success codes are not used by NTRIP. Notably different from common http is that a successful NTRIP Server POST request also receives a 200 OK reply (and not a ‘201 Created’ reply).

Redirection messages

NTRIP does not at this time require support for redirection messages in either Version 1 or Version 2. Some NTRIP Casters may implement it. It is anticipated this may be added as a part of a future NTRIP update.

Client Error Responses

In NTRIP Version 2 several common error messages are supported. A table of these is provided in the next section. **It is extremely important that NTRIP Clients process these messages and correct any problems, and not automatically reconnect to a Caster once one of them is received.** Many such reconnections in a row will often lead to the NTRIP Client device being banned from that Caster.

Server Error Responses

In NTRIP Version 2 three server error codes are defined as follows. The following server error responses are supported but are not commonly used. Note that the Caster may also be located behind a proxy which could provide other replies.

Code / Formal Name	Use in NTRIP
500 Internal Server Error	Nonspecific internal error
501 Not Implemented	Request function / service not implemented on this Caster
503 Service Unavailable	May be used during bandwidth overload conditions

On Standard Caster Error messages

While NTRIP Casters vary in the precise way they reply to unsuccessful connection attempts, the following table lists the common 4xx replies and their meanings. When an NTRIP Client device receives one of these as the returned message, it should correct the indicated error before connecting again.

The following error responses are supported and are commonly used. In each of these cases, additional details about why the request was rejected may be present in the body of the reply.

Code / Formal Name	Use in NTRIP
400 Bad Request	General purpose error code when connection request was not successful.
401 Unauthorized	The NTRIP Client credentials presented are not allowed to connect to the requested resource (Mountpoint).
403 Forbidden	The NTRIP Client request is not allowed, various reasons may be given in the message body (Mountpoint not present, Bad User credentials, outside service area, etc.). Some NTRIP Casters may reply only with 401 and provide no further details.
404 Not Found	The NTRIP Client requested a Mountpoint that is not present on the Caster at this time, or that the requester may not access.
409 Conflict	The NTRIP standard defines this as: Mountpoint is already in use by another NTRIP Server. [Only found with the POST verb used by Base Stations (NTRIP Servers) to connect.]
429 Too Many Requests	May be returned when an NTRIP Client (or NTRIP Server) repeatedly connects to the Caster with incorrect data. May also be returned if the NTRIP Caster is overloaded. May also be returned if the NTRIP Client is connected to the caster simultaneously more than is allowed by the caster.

It should be noted that a large number of well-defined 4xx codes used in other types of web servers are not generally used or supported in NTRIP.

The reply from the Caster will include a complete http header, not just the error response code. It is similar to the replies outlined in other parts of this document. An arbitrary number of other lines containing additional meta data can be present in the header. And these can be present in any ordering. The end of the header is indicated by the final blank line. This is generally followed by a single line providing further details. A representative header is shown below.

```

HTTP/1.1 403 Forbidden<CR><LF>
Date: Valid-UTC-Date-Time-Here<CR><LF>
Server: NTRIP_Caster_Name_String<CR><LF>
Connection: Closed<CR><LF><CR><LF>
This IP or mountpoint has been banned from connecting<CR><LF><CR><LF>

```

In the above example, the text “Forbidden” is considered a keyword associated with the error and will not change. The text line (This IP or ... connecting.) is considered the message and may vary (or be absent) depending on the conditions.

The *further details* will typically be plain text (in standard ASCII) and should be shown to the device user if at all possible. NTRIP uses this text field to convey additional useful information; the specific details will vary with each Caster. Some Casters will send a more complex reply, often starting with <html> <head> and containing other html markup. Some Casters will use HTTP/1.0 rather than HTTP/1.1 in the reply, still others use HTTP/0.9. Some Casters will use the outdated http X-headers as a means to convey errors.

Remarks on Error Conditions that develop when / once a connection has been established

As a general rule, once a connection has been established between an NTRIP Client and a Base station, data flows from the base to the client until the client disconnects. Less often, the connection can be disconnected due to a problem with the Base station (such as loss of connection to the NTRIP Caster) or due to various business rules implemented by the NTRIP Caster (such as the NTRIP Client’s allowed connection time expiring).

An NTRIP Client can also be disconnected due to changes in its operational parameters. For example, an NTRIP Client which connects to a Network RTCM stream (where the specific data stream returned is determined by the reported location of the NTRIP Client) may move to a new location that is not serviced (or cannot be serviced) by that Caster. The behavior of an NTRIP Caster under such conditions is not defined by the NTRIP standard. Generally, such connections are dropped and the NTRIP Client is unable to establish a new connection. At this time there is no mechanism defined in the NTRIP protocol to inform the NTRIP Client of such an event.

On the Use of TLS/SSL

A growing number of NTRIP Casters (and NTRIP Clients) now support the use of secure connections using TLS. This feature has been part of the NTRIP Version 2 standard since its adoption, but deployment has lagged until increased awareness of internet security resulted in increased market demand for this ability. The primary benefits of using TLS connections in NTRIP are twofold;

The connection and data source itself is authenticated in that the owner of the security certificate (the “CERT”) is the from the organization and/or the URL/IP that it claims to be from.

The data stream itself has not been corrupted or manipulated by any intermediary devices that carried the message content (the “man in the middle” attack which could spoof correctional data resulting in incorrect navigational results).

A TLS connection is similar to any other TCP/IP connection. After the initial negotiation process completes, the data is secured using a block encoding⁸ method, encoding raw data with the private key of the sender and decoding it in the NTRIP Client with a public key that is included in the CERT which is exchanged during the negotiation process. At the start of the TLS connection the NTRIP Caster presents the CERT it will use, and this in turn lists a number of cryptographic methods which it supports. The NTRIP Client will examine the CERT for various possible defects and then select suitable cryptographic methods, and both parties will agree to use it.

From the application programming level (NTRIP Client software development) much of the process above occurs at lower levels and is transparent. The application makes a connection to the remote host and the initial connection negotiation process completes (passing or failing) without further involvement. In the case of a successful connection, the decrypted data is presented from the TCP/IP socket to the application layer in the same fashion as a non-secure connection. The only additional programming effort required is how to handle CERT that are not considered to be valid⁹ or were not issued by a known CA (considered a *self-signed* CERT). When such an event occurs, the NTRIP Client software needs to present the details about the event to the user to allow them to make a determination regarding what to do. In the case of *self-signed* CERTs, or a CERT that cannot be traced back to a root CA, is it common to present the CERT details to the user and ask if they are willing to accept the CERT. Note that using *self-signed* CERTs for financial transactions is not advised, but for RTK connections where the risk of loss is very low, it may be practical and pragmatic to do so. It should also be noted that a *self-signed* CERT does not provide any greater or lesser level of security for the connection than a similar CERT issued by a CA. Most developers use widely available toolkits, such as OpenSSL¹⁰, to provide the basic cryptographic services they need within their preferred development environment.

Typically, a secure connection for an NTRIP Caster will be operated on its own port. The port 443 is commonly used for secure http web casters (an https service), but if the host machine services both general web requests and secure NTRIP requests, they often serve NTRIP requests on a different port. NTRIP makes no recommendation in this regard. It is not unusual for a Caster to operate both a non-secure (traditional) port and a secure one. At this time there is no redirect mechanism in NTRIP to inform the NTRIP Client that a secure alternative exists. As a general recommendation, use a secure connection when one is available, and develop your NTRIP Client software with this in mind, as adoption and use of secure Casters is expected to become the industry norm.

⁸ An important side effect of this is that the payload size of RTK data itself is left unchanged. So bandwidth limited connections are not stressed when using a secure connection.

⁹ A CERT may be invalid for several reasons, including not being associated with the IP or URL name it gives – reasons which would normally be a cause for rejecting it. A CERT may also be invalid because it has expired (modern CERTs are valid for about one year after being issued). A CERT may also be invalid if it was not issued by a trusted CA, which in many instances may not be a problem.

¹⁰ For further details see <https://www.openssl.org>

Concluding Remarks

This document was developed by the NTRIP Working Group of RTCM SC-104 to assist parties developing NTRIP Clients. It is our goal and expectation that the information contained in this document will enable developers to create better and more robust NTRIP Clients. Additional detailed technical information about NTRIP can be found in the standards themselves, and the reader is urged to become familiar with these, particularly the most recent version. They are available from www.RTCM.org

RTCM 10410.0 (RTCM Paper 200-2004/SC104-STD, Version 1.0), with Amendment 1,
Standard for Networked Transport of RTCM via Internet Protocol (Ntrip), September 30, 2004

RTCM 10410.1 Standard for Networked Transport of RTCM via Internet Protocol (Ntrip)
Version 2.0 with Amendment 2, January 12, 2021

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